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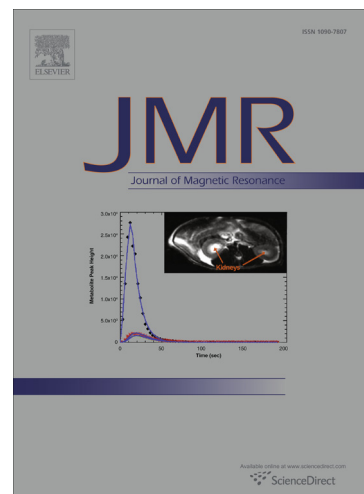
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Symmetrically Biased T/R Switches for NMR and MRI with Microsecond Dead Time

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Highlights

- A novel PIN diode based T/R switch topology is introduced.
- Switching below 1 μ s is demonstrated by bench and MRI measurements.
- Very low spurious transient signals are achieved.
- Enhanced switching performance is achieved with state-of-the art signal efficiencies and linearity.

Abstract

For direct NMR detection and imaging of compounds with very short coherence life times the dead time between radio-frequency (RF) pulse and reception of the free induction decay (FID) is a major limiting factor. It is typically dominated by the transient and recovery times of currently available transmit-receive (T/R) switches and amplification chains.

A novel PIN diode-based T/R switch topology is introduced allowing for fast switching by high bias transient currents but nevertheless producing a very low video leakage signal and insertion loss (0.5 dB). The low transient spike level in conjunction with the high isolation (75 dB) prevent saturation of the preamplifier entirely which consequently does not require time for recovery.

Switching between transmission and reception is demonstrated within less than 1 μ s in bench tests as well as in acquisitions of FIDs and zero echo time (ZTE) images with bandwidths up to 500 kHz at 7 T. Thereby the 2 kW switch exhibited a rise-time of 350 ns (10%-99%) producing however a total video leakage of below 20 mV peak-to-peak and less than -89 dBm in-band.

The achieved switching time renders the RF pulse itself the dominant contribution to the dead time in which a coherence cannot be observed, thus making pulsed NMR experiments almost time-optimal even for compounds with very short signal life times.

Introduction

Transmit-receive (T/R) switches are integral parts of pulsed NMR and MRI systems enabling time-interleaved connection to the transmission and reception chain for the same probe port. High isolation protects the sensitive low noise components of the receivers during high power radio-frequency (RF) pulses applied for spin nutation. This renders the switches the key element deployed for mastering the extremely high dynamic range between the applied short transmission pulses -reaching several kilowatts - and the received signal below a microwatt. The key figures of an RF switch are the provided isolation, the introduced losses diminishing the system's transmit efficiency and noise performance, its linearity under high power pulses, the provided switching speed, and the concomitant spurious signals on the RF signal lines (video leakage) [1]. These figures are subject to stringent technical trade-offs and hence switches need to be tailored to the particular application. Switches deploying positive-intrinsic-negative (PIN) diodes as the active elements are to date ubiquitous in NMR and MRI. This is because the diodes offer very low on-state resistance, high off-state impedance, high linearity and power handling capabilities, as well as high reliability and false state detectability. Consequently alternative technologies based on transistors [2, 3] or micro-electro-mechanical systems (MEMS) [4] still have not found broad application in MR despite their prospering in other areas.

Usually, the dead time between transmission and reception phases incurred by current systems is not critically hindering NMR and MRI experiments because the typically observed spin coherences have long life times, and time intervals for spatial or spectral encoding of the signal are required ahead of data acquisition anyway. Even for NMR of solids, relatively large gaps can be accepted when broad spectral lines are narrowed by averaging out dipolar couplings by means of magic angle spinning (MAS). However, MAS is typically not applied for imaging of compounds with very short transverse relaxation times such as solids, polymers, bone, myelin, bound water pools, etc. Furthermore, rapid spinning is obviously not an option for most biological and in-vivo samples. Alternatively, the effect of spectral broadening on image resolution can be counteracted by rapid data acquisition under strong encoding gradients. This approach is employed for imaging compounds with very rapid signal loss (<1 ms) by techniques with ultra-short [5, 6] and zero echo time [7-11] which, however, both require very short dead times, thus pushing the demands for faster switching. In many cases the benefit of decreasing the dead time even scales beyond reduced signal loss during T/R switching, namely by enabling larger acquisition bandwidth, improving the numerical condition of the reconstruction problem [10, 12], reducing the total acquisition time or improving signal consistency. Also, for the acquisition of spectra of quadrupolar nuclei [13] or baseline corrections [14, 15], short dead times are of critical importance for preventing irrecoverable loss of information induced by relaxation.

The receiver dead time after transmission is composed of the coil ring-down time, the signal rise time and in-band noise transient decay time of the switch, in addition to the recovery time of the receiver chain as well as the lead-in time of the applied signal filters in digital and analogue domain. While the relatively low quality factors of loaded coils applied for in-vivo imaging often offer a comparably short ring-down, the latter can also be actively suppressed [16], subtracted [17] by additional circuitry or avoided by tailored pulse waveforms [18-20] resulting in delays on the order of 100 ns for preventing the switch being thrown under significant RF power load (hot switching). Furthermore, the lead-in time of band-pass filters in the receive chain was shown to be reducible to the same order of magnitude using broad-band receivers and variable-rate digital down conversions, with little impact on the requirements for the data throughput and storage [21]. Therefore, for transmit-receive coils and arrays, the transient time of the T/R switch and of the analogue receive chain to settle and to choke spurious signals generated by the control voltage transients nowadays remains the major contribution to the dead time. However, the reported minimum switching times scale greatly with the requirements set to insertion loss, linearity and power handling. For low power applications as driving micro coils they are found to be in the order of 1 μ s [22]. But for power levels in the range of kilowatts switching times are as short as of 4.5 μ s in typical high resolution spectrometers and animal imagers [10] and 8 μ s or more on human MRI scanners [23, 24], limiting the above mentioned methods. Although continuous wave NMR does in principle provide zero dead time, and attempts to measure the signal concurrently with its excitation have been reported in MRI and MAS settings [25-28], the impact of these methods on applicable power levels, power consumption, speed, isolation requirements, system complexity, or robustness to coil-load variation have prevented their broad application to date.

Increasing the switching speed in available PIN diode T/R switch topologies by simply using faster diodes or reducing the forward biasing negatively impacts the resulting RF performance in terms of linearity of power signal transmission, signal losses and isolation [29]. Furthermore, when switching very fast by applying higher and faster rising reverse voltages, the video leakage signal generated by the transients of the diode biasing signals can saturate or affect the bias point of the preamplifier, which then requires a recovery time – typically of the order of a few μ s [30] to a few tens of μ s [31] - in order to provide a stable and constant signal transmission. In fact, the total signal transient time of NMR systems employing fast PIN diode switches are reported to be dominated by the recovery time of the preamplifier [32]. This video leakage can also be filtered only moderately, because the signals required to switch and settle within less than 1 μ s reside at frequencies comparably close to the frequency of the RF carrier. Also the bias lines cannot offer sufficient bandwidth to provide very fast biasing signal transients, due to the large inductance of the RF chokes required to block the RF signals on the diodes.

In this work we introduce a novel topology that uses opposite signalling modes for the RF signals and the biasing of the PIN diodes [33]. This allows separation of the two signal pathways by their modal structure, which greatly suppresses the video leakage. Furthermore, the RF can be isolated from the bias lines by common-mode chokes without introducing a large RF-blocking inductance into the bias signal path. Thereby the PIN diodes can be biased with high-bandwidth signals. This allows throwing a highly linear switch in submicroseconds while maintaining the video leakage on a very low level. In conjunction with the provided isolation, the preamplifier is prevented from being saturated at any stage of the experiment resulting in a total dead time due to switching of less than 1 μ s. The novel topology is implemented, tested and used on a 7 T whole body system for zero echo time (ZTE) imaging with bandwidths up to 500 kHz using peak powers up to 2 kW per channel.

Methods

RF circuitry

For application in MRI, the low voltage configuration of a PIN diode T/R switch (Figure 1a) is typically preferred if the unit has to enter the magnet bore because it does not require very high bias voltages to sustain reverse polarization of the diode during the RF pulse. During transmission, the diodes are forward-biased by providing a DC current (I_F) through the bias choke. In proportion to I_F and the intrinsic charge carrier life time (τ ; the time within the charges would recombine at no external biasing), charges accumulate within the PIN diode. These charges render a high degree of conductivity by which the impedance between the diode's terminals becomes very low for the RF current. This connects the power amplifier wired to the transmit port (TX) to the coil at transmit-receive (TR) port, while the digitizer at the receive (RX) port is isolated via a 90° delay line ($\lambda/4$). In this state, linearity of the signal transmission through the diodes directly depends on the amount of the accumulated charges in their intrinsic layer if the period of an RF cycle is much shorter than τ [34]. For throwing the switch back to receive operation this charge has to be removed from the intrinsic layer. This happens passively by the recombination process in the diode itself and can be driven actively by applying a reverse voltage leading to the transient reverse current of these intrinsic charges (I_R) from the diode. Consequently the time for the recovery from forward to reverse bias state (T_{FR}) for large I_R is given by [35]:

$$T_{FR} \approx \tau \cdot \ln \left(1 + \frac{I_F}{I_R} \right) \quad [1]$$

Hence this recovery time can be reduced by deploying PIN diodes with shorter charge carrier life times, reducing the forward bias current and increasing the reverse transient bias current and voltages. While the first two measures have a direct impact on the linearity of the RF signal transmission, power handling and losses induced by the switch, increasing the reverse voltage amplifies signal spikes and spurious signals on the RF lines in reaction to the required faster slew rate. Hence the inherent properties of PIN diodes set clear trade-offs between switching speed, video leakage, linearity, isolation and losses when designing switches in the conventional topology of Fig. 1a.

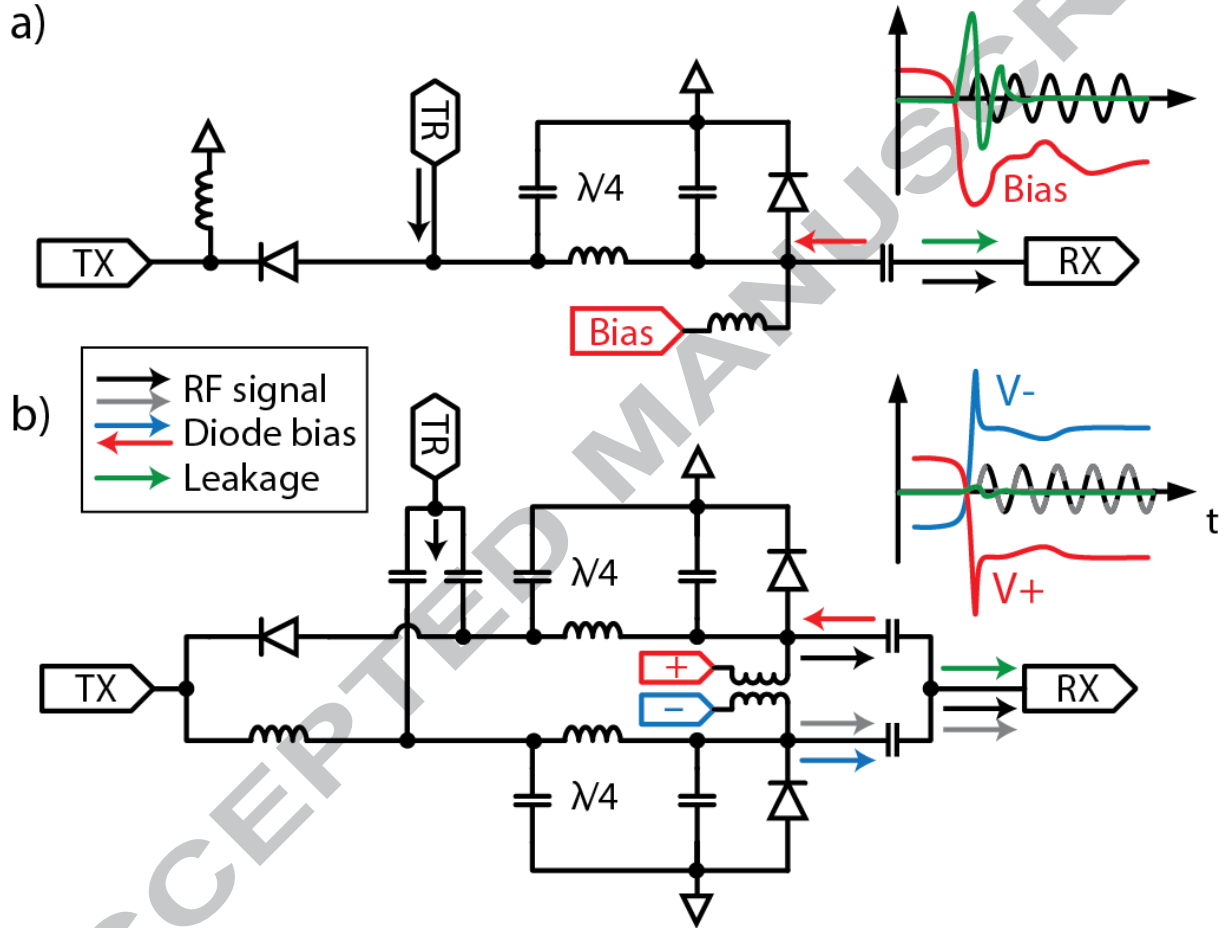


Figure 1: a) Basic schematic of a PIN diode T/R-switch with low bias voltage topology connecting the coil port (TR) either to the transmitter (TX) or to the receiver (RX). The slewing of the bias voltage (red) is hampered by the bias choke required to block the RF signals from the diode driver. Furthermore a transient spike signal (green) is added to the RF receive signals (black) creating a large video leakage signal when switching fast. b) Novel approach using differential diode biasing and single ended RF signals. This inherently cancels the video leakage on the RF ports and allows using common-mode chokes for blocking the RF signal flowing in common-mode on the two centre lines without introducing a large

inductance on the biasing signal pathways running in the differential mode to the PIN diodes through the transformers.

To circumvent these relations, the basic topology of a low voltage, lumped element, high power PIN diode T/R switch (Fig. 1a) is extended by splitting the center conductor of the single-ended RF signal transmission lines and polarizing the PIN diodes of the RX isolation stages in an anti-parallel configuration (Fig. 1b, [33]). The biasing to the diodes is provided by common-mode chokes made from bifilar toroidal inductors (inner/outer diameter: 4/12 mm). Since the RF signal is guided single-ended, the high common-mode impedance of these transformers blocks the RF from the PIN diode driver. Yet, for the biasing currents being routed differentially, the common-mode choke exhibits a very low resistance and inductance, thus allowing high transient currents and broadband signals to be provided to the diodes without high excess voltage spikes being produced across the choke. The TX line is isolated from the RX signal path during reception by the high impedance provided by a reverse-biased PIN diode forming a tank circuit on the two center conductors with the diode's capacitance and the bias inductor. During transmission, this diode is forward-biased, bypasses the tank circuit and connects the TX port to the antenna. The biasing to this tank circuit is provided differentially and thereby isolated from the RF mode.

The schematic of the actual implementation is given in Fig. 2. In order to reduce unwanted common mode current return paths to the diode driver over ground and to reduce the voltage swing on each diode terminal, the RF ground of the switch is AC coupled to the ground of the diode driver. The bias currents are applied to the diodes via two separate bifilar, toroidal common mode chokes (Figure 2b). To avoid full saturation of the employed low noise pHEMT preamplifier under 2 kW of transmitted power, 66 dB of isolation has to be provided. This was achieved by two consecutive 50 Ω quarter-wave sections made of lumped element PI structures. The impedance match of the RF ports of the switch were fine-tuned by individual T-section circuits which further block low frequency signals from and to the switch circuitry. Furthermore, the grounds of the ports of the switch from which the cables are routed out of the bore (TX, RX) are AC coupled in order to avoid induction of ground currents into the cables by gradient switching.

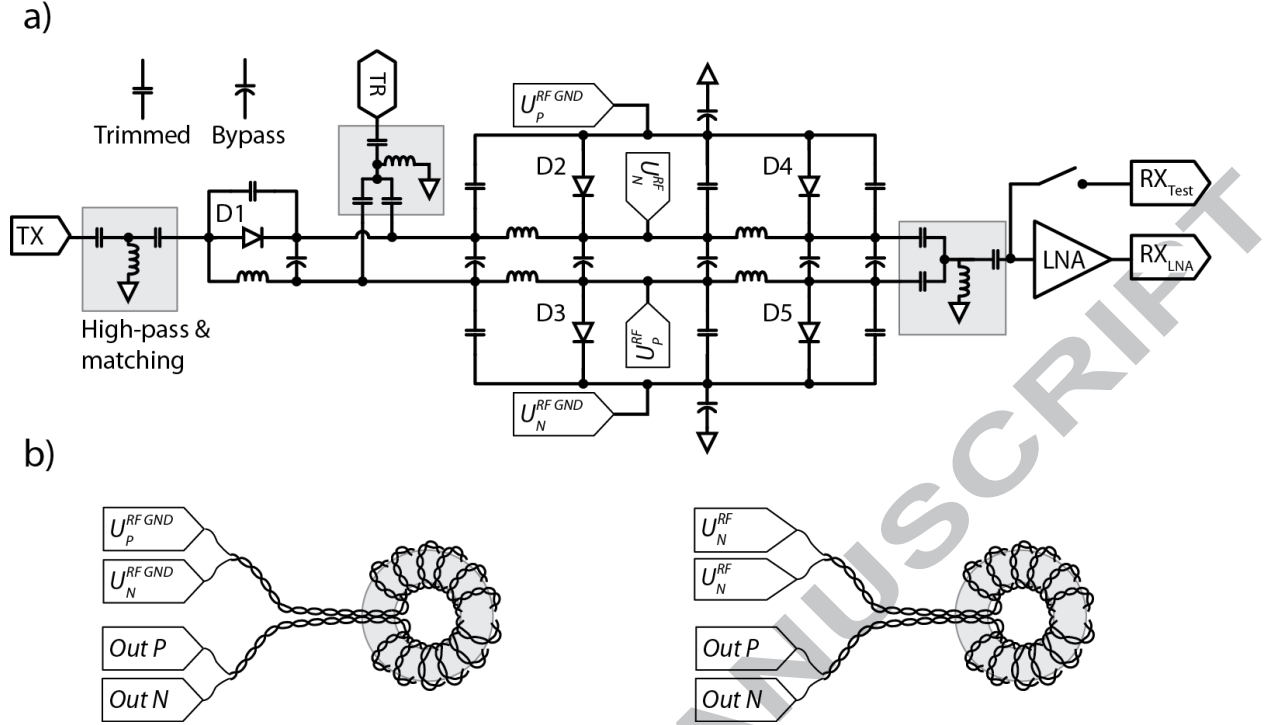


Figure 2: Schematic of the RF signal paths of the proposed T/R-switch (a) and the bifilar common-mode chokes (b). The differential ports $U_{P,N}^{RF}$, $U_{P,N}^{RF GND}$ are each connected to a common mode choke. The RF ports (TX, TR, RX) are connected via T-sections (grey boxes) for fine tuning impedance matching and providing additional high-pass filtering (b) for isolating the RF signals from the bias lines. D1 ... D5 denote the RF switching PIN diodes (MA4P505-1072T, MACOM, Lowell, USA).

PIN diode driver

Unlike in traditional T/R switch designs with single-ended PIN diode biasing, a driver that provides high differential output currents and voltages is required. Furthermore, the driver must ensure a very high directivity into the differential mode, because the video leakage on the RF lines is driven by the common-mode contributions of the PIN diode biasing transients and not by the biasing transient on the diode itself. The bias driver (Fig. 3a) was implemented with active feedback regulation of the bias voltages [36] in order to provide a quick settling of the forward and reverse bias voltages and to suppress bias oscillations which would result in unwanted RF signal modulations after throwing the switch. The device employs a differential operational amplifier (U2), regulating a bipolar, differential, push-pull, class B output stage and is designed to provide 1.8 A of peak current and a differential reverse voltage of up to 30 V. The switching signal provided to the driver is a standard logic low-voltage transistor-transistor logic (TTL); however, an additional operational amplifier (U1) circuit is employed for level shifting and preconditioning of the slope of the switching transient in order to provide a bandwidth-controlled voltage

slew as an input to the diode driver. By this circuit the forward bias (100 mA in total) as well as the reverse voltage (22 V) can be adjusted.

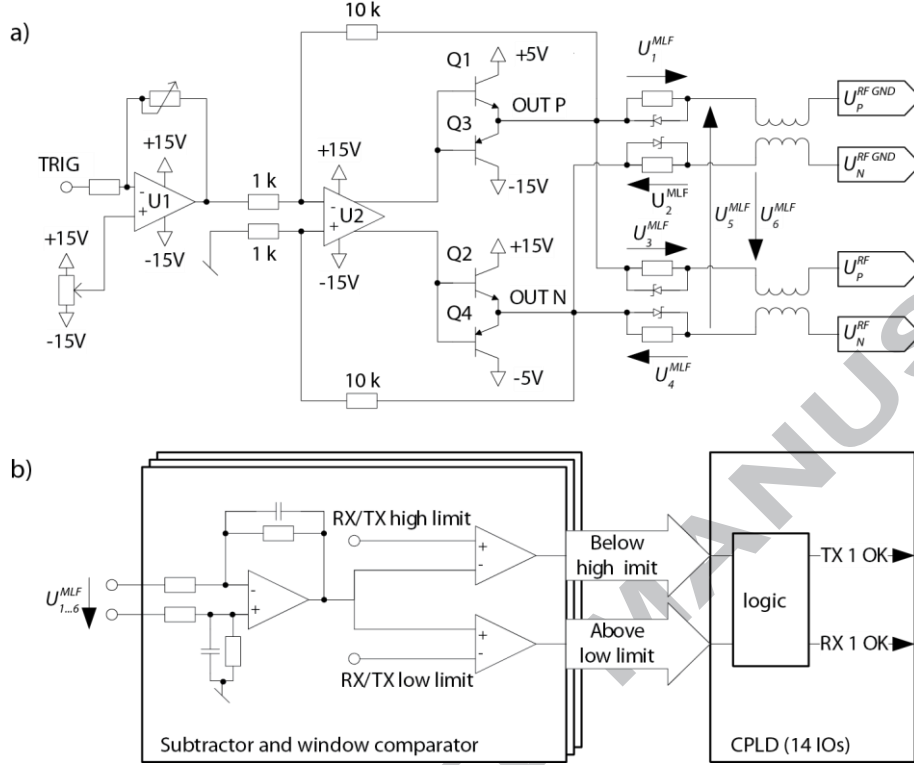


Figure 3: a) Simplified schematic of the feedback-regulated, high peak-current, differential PIN diode driver employed. The operational amplifier U1 (THS4011, Texas Instruments, Austin, USA) is used for level shifting and pulse shaping and U2 (THS4130, Texas Instruments, Austin, USA) for feedback control of the output stage. Each of the push-pull output stages consists of a matched pair (ZXTC2063E6, Zetex, Plano, USA) of NPN (Q1, Q2) and PNP (Q3, Q4) transistors b) Block diagram of the malfunction detection circuit implemented to monitor the biasing currents and voltages of the diodes to determine the integrity of the diodes in real time during operation.

Malfunction detection

For safety reasons and to avoid destruction of the receive chain, the integrity of the T/R switch is assessed in real time by checking on the bias conditions of the PIN diodes (Fig. 3b). Hence a failure or significant degradation of one of the diodes would be sensed by an alteration of its biasing current and reverse voltage. The bias current is measured by the voltage drop (U_i^{MLF} , $i = 1 \dots 4$) over four bias resistors (10 Ω) which are bypassed for the reverse bias current by Schottky diodes. The reverse voltage (U_i^{MLF} , $i = 5, 6$) is differentially sensed over the bias lines. These voltages are checked by individual window comparators,

setting the tolerance under forward and reverse bias condition. The windowing signals are then individually debounced by monostable timing circuits with an RC time constant of 330 ns and combined in a complex programmable logic device (CPLD) to provide a single state-assertion logic-signal confirming the safe latching of the requested state.

Implementation and Integration

The RF circuitry, the diode driver and its malfunction detection are deployed on a single board (FR4, class 6, 6 layers) module. A motherboard able to hold, supply, and control 8 of these units in parallel by means of a CPLD is constructed from a single PCB (FR4, class 6, 6 layers). This mother board connects to the power supply and control unit located outside the Faraday cage. While the supply (± 15 , ± 5 , DC Ref GND) is delivered by a twisted-pair, multi-strand copper cable, the sensitive real time trigger and malfunction signals are routed via two unidirectional optical fibers for ensuring electromagnetic immunity. The entire unit exhibits a low magnetic moment and was placed in the bore of the MR system for the experiments. The external supply unit consisted of 4 double isolated AC/DC power supply units providing the required voltages from a standard mains connector (230 V, 50 Hz), a power sequencing circuitry avoiding latching scenarios during start-up, power good assertion, and the electro-optical converters interfacing to standard TTL. A full block diagram of the system is provided in Fig. 4.

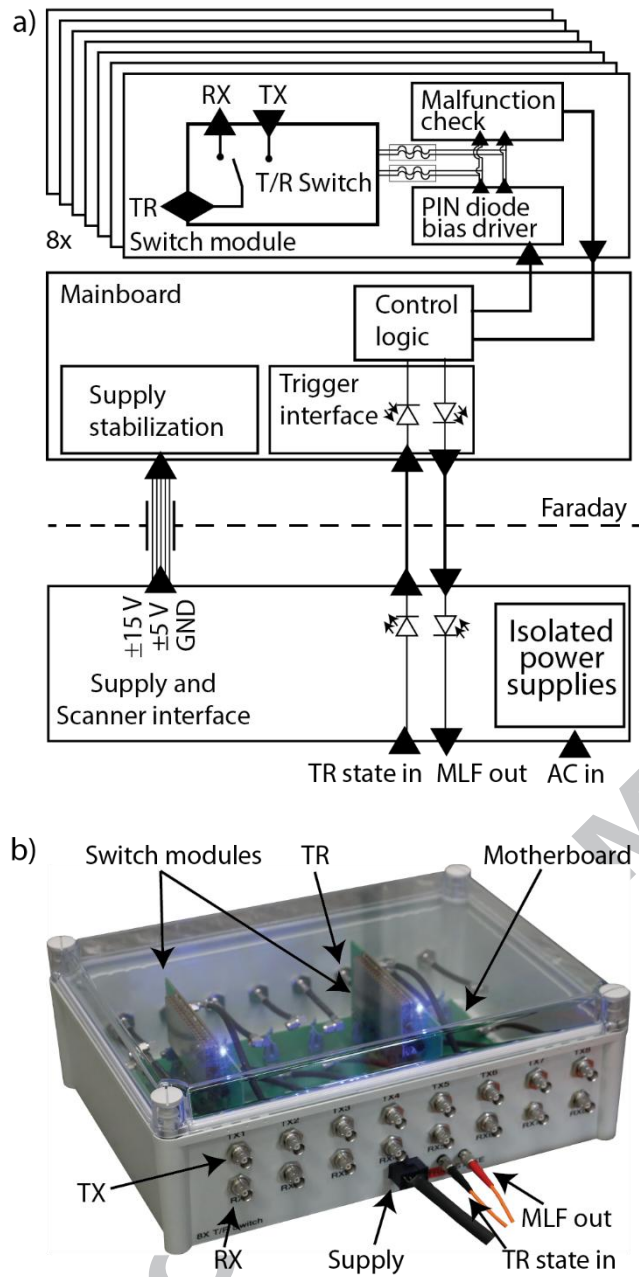


Figure 4: a) Block diagram of the integration of 8 T/R-switch modules for driving parallel transmit arrays, including the supply and scanner interface module located outside the Faraday cage. A motherboard stabilizes and distributes the supply voltages and trigger signals for synchronously throwing the switches triggered via a fiber optic interface. The malfunction signals of occupied slots are logically combined to a single optical output (MLF out) indicating that all switches have safely latched the requested state. b) Photograph of the switching unit hosting 2 T/R switch boards.

Bench testing

Bench tests were performed using the equipment as listed in Table 1. For assessing the performance of the switch alone, an additional, coaxial RX port was used for bypassing the preamplifier. Insertion loss and matching condition of all ports in TX and RX state were measured by a calibrated vector network analyzer.

Switching transients of the diode biasing were recorded using a mixed signal oscilloscope with high impedance inputs (2 M Ω). The measurement tips were referenced to the ground of the unit, and acquired synchronously. The differential voltages were then derived numerically using the relations set in Table 2. The effective bias voltages on the terminations of the PIN diodes on both sides of the differential structure ($U_i^{Bias}, i = 1,2$) were assessed for showing their relation to the transient of the RF signal transmission. Furthermore, the signal transients on the single-ended center line of the RF mode ($U_{RF\ signal}^{CM}$) and its ground ($U_{RF\ GND}^{CM}$) were determined by the corresponding common-mode voltages during switching for estimating the video leakage inside the switch. The output video leakage of the switch was finally determined on the RX_{Test} port by terminating the oscilloscope's input to 50 Ω as well as by calculating the voltage of the RF mode inside the switch (U_{RF}^{SE}).

The transients of the RF signals were measured with the preamplifier in the signal path. The in-band spurious signal was measured with terminated antenna (TR) and TX ports and by recording the preamplifier's output using the spectrum analyzer's internal signal demodulator board (bandwidth: 18 MHz). The signal rise time was assessed in the same setup by feeding a center frequency carrier at -50 dBm from the signal generator to the TR port and recording the resulting signal modulation after the preamplifier. The rise time was then determined by the time required to rise the RF signal from 10% to 99% of the full signal level at the RX output.

The power handling and linearity of the switch was assessed using a stand-alone 1 kW power amplifier and the signal generator in pulsed mode providing pulses of 2 μ s duration. The forward output power of the power amplifier was regulated by attenuation of the input using a step attenuator and tapped to the spectrum analyzer by a directional coupler before a dump load. In a first step, the output power curve of the amplifier was determined. Then the T/R switch under test was inserted into the signal path in-between the amplifier's output and the directional coupler. The compression curve of the switch was then determined by comparing the output powers of the switch and the amplifier alone which was measured separately using the same settings. Furthermore, the signal power during the pulse at the 2nd (596 MHz) and 3rd (894 MHz) harmonic overtone was measured with and without the switch in the signal path.

Imaging experiments

The switch was tested on a 7 T whole body human scanner using the general setup described in Ref. [37]. The RF TX and RX chains were provided by a stand-alone acquisition system synchronized with the MRI scanner via a TTL trigger line. Block-shaped RF pulses of 2 μ s duration were produced by a pulse-modulated signal generator timed by a TTL trigger controlled by the acquisition system. The pulses were sent to the power amplifier, using up to 1.8 kW for flip angles of about 2°, and finally routed to the T/R switch. The acquisition system further provides a second, fast, high-isolation but low-power signal switch at the input of its gain stages that was employed during the transmission phase to prevent compression of the digitizer's analogue stages. The relative timing of pulse generation, T/R switch operation, second stage switch, and blanking of the power amplifier was provided by the acquisition system and was fine adjusted in relative steps of 100 ns in order to provide the shortest signal transient when connected to the 50 Ω terminated oscilloscope. In particular, the timing of the T/R switch had to be adjusted relative to the falling slope of the RF power in the switch caused by the coil ring-down in order to prevent hot-switching conditions. By doing so, the system was implicitly adjusted for propagation delays of the various instruments.

Two coils were employed, a custom made, single-port 10 cm square loop coil with low proton content, driven by the Bruker amplifier, and a standard quadrature head volume T/R resonator, driven by the 4 kW amplifier. The volume coil was fed in quadrature via a hybrid and connected with two separate T/R switches on both ports of the coil. FIDs and high bandwidth ZTE images were acquired from a bottle filled with mineral oil (Marcol 82, ExxonMobil, Irving, USA), a piece of rubber mat, and the head of a healthy volunteer in accordance with the local ethical approval. Since the volume resonator is housed in plastic material giving proton signal of short T_2 , the field of view had to incorporate the entire coil resulting in very high imaging bandwidths. Data acquisition was performed at 1 MHz output bandwidth, and images were reconstructed at 250 and 500 kHz, resulting in oversampling factors of 4 and 2, respectively, as required by finite-support based algebraic reconstruction [38]. The length of the applied digital filter was 4 μ s. In total, the dead time amounted to 4 μ s, corresponding to 1 or 2 Nyquist dwells for the different signal bandwidths. For comparison, ZTE images with the highest bandwidth were acquired in an identical setup using custom-built, T/R switches with conventional single-ended biasing (unpublished) offering signal transient times of about 3.5 μ s. The relative timings of the pulse generator and the switches were equally optimized as described above.

Results

Table 3 collects the results from the S-parameter measurements in both states showing that good impedance matching conditions could be achieved for all active ports as well as low insertion losses were realized for the transmission and reception path.

Figure 5a shows the voltage transients relative to the input trigger for the critical switching from transmit to receive without the preamplifier in the system. It is observed that 400 ns after the trigger-slope the final differential reverse bias is applied and stable. Although the bias is slewed 22 V within 300 ns, the signal spikes on the RF mode measured at RX_{Test} are only 20 mV peak-to-peak. The corresponding measurement with the preamplifier in the chain (Fig. 5b) shows that the RF rise time is equally as fast as the slewing of the PIN diode biasing suggests (350 ns for 10%-99% rise). Also with preamplifier, the leakage to the RF common modes ($U_{RF\ signal}^{CM}$, $U_{RF\ GND}^{CM}$), the RF signal mode (U_{RF}^{SE}) as well as the input of the preamplifier (U_{RX}) remain on very low levels. Furthermore, the signal transmission is found to be stable in amplitude and phase (not plotted) afterwards. Similarly, the in-band spurious signals exhibit an extremely low level of -89 dBm when subtracting the gain of the preamplifier, and drop below measureable levels within the switching transient time. The slight enhancement of the noise floor found in receive state compared to the transmit state is related to the change in input impedance presented to the preamplifier which offers its peak performance at the 50 Ω seen during reception.

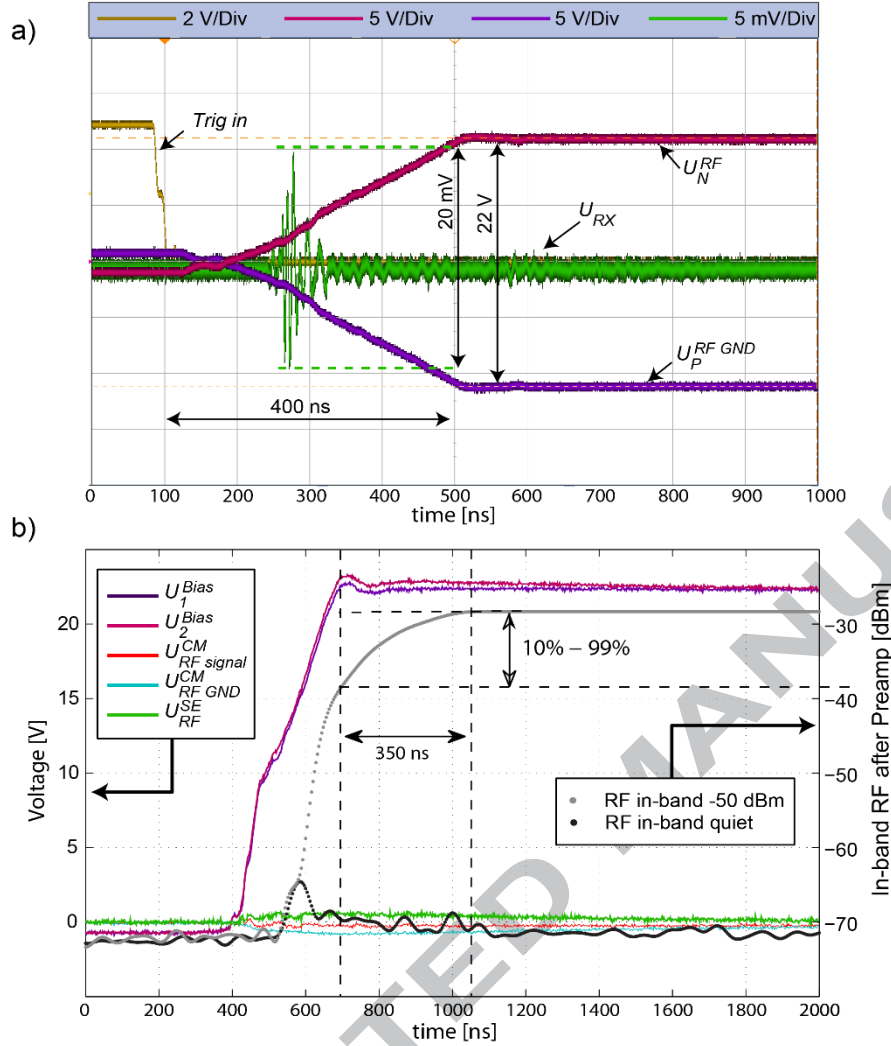


Figure 5: Bench results. a) Transient voltages from switching of the input trigger (Trig in), the bias voltages at the terminals of the PIN diodes (U_N^{RF} , $U_P^{RF GND}$) and at the RX_{Test} port without preamplifier (U_{RX}) measured relative to GND. b) Differential voltage transients measured with preamplifier (RX_{LNA}). Furthermore, the in-band (298 MHz) signal rise time and RF leakage are plotted including the gain of the preamplifier.

The linearity measurements in Fig. 6 show that the switch is highly linear with a deviation on the order of 0.1 dB up to the 60 dBm that could be provided using the stand-alone power amplifier. Furthermore, it is assumed that the found compression is related to alterations of the linearity of the amplifier itself due to slight changes of the output load presented during the calibration and the measurement setup. This assumption is supported by the measurements of the higher harmonics which showed virtually no change due to the insertion of the switch being 13 dBm and 6.3 dBm for the 2nd and 3rd harmonic at 61 dBm output power of the amplifier. Also the fact that the apparent insertion loss was found to be 0.5 dB higher throughout all measurable power levels compared to the measurement using the network analyzer is

believed to be related to the amplifier changing transmission properties due to small differences in the output impedance.

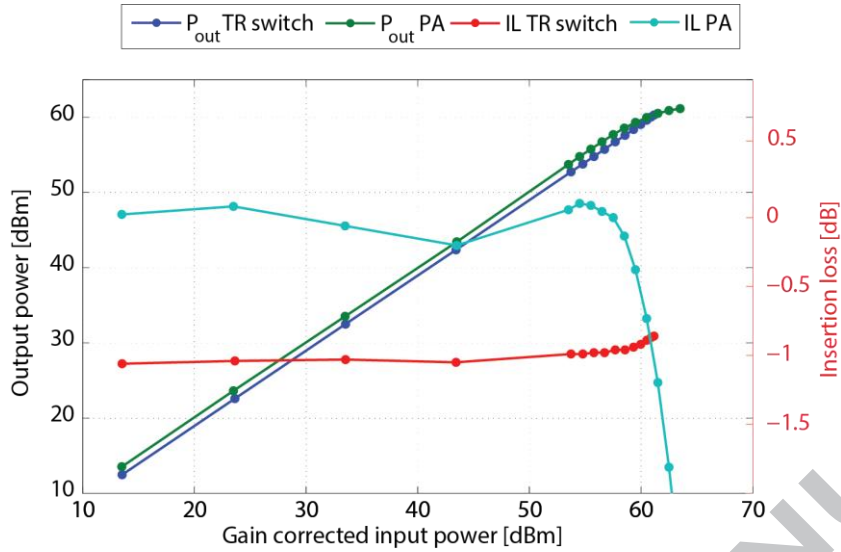


Figure 6: Amplitude compression measurements of the power amplifier alone (PA) and of the T/R-switch correcting for the compression of the amplifier. IL = insertion loss.

Figure 7 collects the results obtained from the 7 T MR measurements. The FIDs (Fig. 7a) acquired from the different samples show no visible distortions or signs of spurious signals after the digital filter lead-in time. Accounting for this effect, the analogue signal could therefore be sampled 1 μ s after the high power RF pulse. Furthermore, the FIDs acquired with the volume coil clearly show the short-lived components of the coil housing with a T_2^* of about 20 μ s on top of coherences with significantly longer signal life times. Figs. 7b) and c) show ZTE scans from the oil bottle using the volume resonator acquired with imaging bandwidths of 250 kHz and 500 kHz. The color scales of the images are levelled to show the signal background. By this, infidelities and spurious signals contaminating the ZTE reconstruction are highlighted and the depiction quality of the obtained images can be assessed best. Even with 500 kHz signal bandwidth the background stays clear and the resolution of short-lived materials such as the phantom support or the coil housing is clearly enhanced by the stronger read gradient versus the lower-bandwidth acquisition. Figure 7d) shows the comparison to a standard single-ended switch. The signal transients lead to a significant haloing around the isocentre of the image and deteriorate the signal localization thereby. Figure 7e) was acquired with the surface coil connected to the novel T/R switch with a bandwidth of 500 kHz and exhibits, besides the expected ringing of the point spread function, no visible artefacts in its background. The same setup was used to acquire an image of a short- T_2 rubber mat shown in Fig. 7f). Figure 7g) shows a transverse slice of a ZTE acquisition of a human head acquired with an imaging bandwidth of 500 kHz using the volume coil.

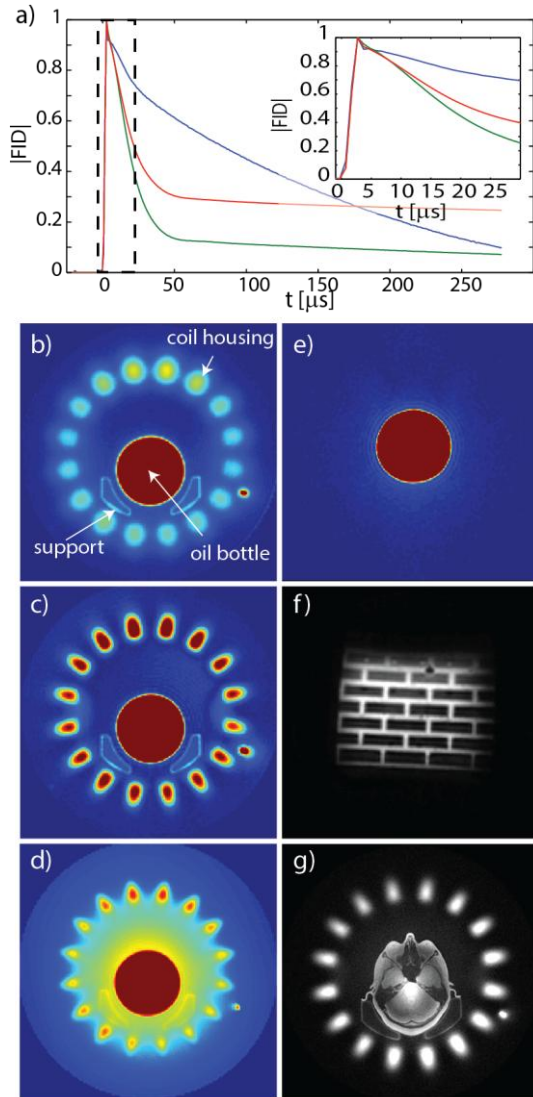


Figure 7: Test measurements of the switch in the 7T MR scanner. a) FIDs acquired from a rubber mat ($T_2^* \approx 220 \mu s$) using the proton-free loop coil (blue), as well as mineral oil (red) and a human head (green) using the volume coil. The time $t = 0$ corresponds to the RF pulse center. b)-d) False color images acquired using ZTE with the volume coil. The images are scaled in order to assess the clarity of the background signal. FOV = 530 mm, matrix = 128. b) and c) were acquired with the proposed switches with 250 kHz and 500 kHz bandwidth, respectively. d) was acquired at 500 kHz with traditionally designed TR switches offering a signal transient time of about $3.5 \mu s$. e) False color image of the same oil bottle at 500 kHz using the low proton content surface coil. FOV = 340 mm, matrix = 128. f) Image of a rubber mat using the surface-coil setup. FOV = 240 mm, matrix = 128. g) ZTE image of a human head in the volume resonator using 500 kHz bandwidth. FOV = 530 mm, matrix = 320. All images were acquired with a repetition time of 0.5 ms and full angular Nyquist encoding, resulting in scan times of 51 s and 2 m 31 s for the 128 and 320 matrices, respectively.

Discussion

The presented design of T/R switches with symmetric PIN diode biasing is shown to enable switching from transmission to reception within 1 μ s with no additional time for recovery of the receive chain. The implemented device is able to hold 8 switches operated in parallel and can be fully integrated into existing NMR and MRI systems via a robust optical trigger and a malfunction detection line. With a power rating of 2 kW, each unit handles the maximum available power at the coil port offered by the current scanner configuration in single, dual and 8-channel transmission configurations. The peak power handling capabilities of the current implementation are mainly limited by the voltage rating of the employed capacitors.

The short dead time could be realized because the new topology inherently cancels the video leakage of the switch and at the same time allows for biasing of the PIN diodes with broadband, low impedance signal lines without large inductances in the biasing circuitry. The resulting high transient biasing current swiftly depletes the charges in the intrinsic layer of the diode, accelerating the reverse recovery of the diode without saturating or endangering the preamplifier. Since the video leakage signal is isolated from the RF signal by the orthogonal symmetry of their signaling mode, the video leakage spike is mainly determined by the differential balance of the diode driver and the common-mode blocking towards the RF signal ports. In the present design, this has been realized by making use of the common-mode rejection of a high-speed differential amplifier regulating the biasing voltage which further suppressed bias line oscillations without a specific tuning of the driver to the presented load [36]. The demonstrated RF performance of the novel switch in terms of losses and linearity was found to be comparable to purely single-ended topologies. Consequently, the boundaries set by the trade-off inherent to traditional topologies between switching speed, video leakage performance, linearity and signal efficiency of the switch can be significantly expanded or even circumvented. Furthermore, the anti-parallel arrangement of the PIN diodes is expected to enhance the linearity performance as such compared to a parallel arrangement [39].

With the demonstrated switching time the total dead time in which a coherence cannot be observed is dominated by the RF pulse itself given that pulse lengths of at least a few μ s are required to achieve a sufficiently large flip angle of a few degrees. This renders the time efficiency of a time-interleaved, pulsed NMR experiments close to optimal in the sense that practically all the experimental time can be used for either transmission or reception even in the case of very short pulsing and rapid repetition rates. The provided dead-times not only allow for direct detection of NMR coherences with life times in the microsecond range but ZTE- and SWIF type sequences can be applied with unprecedented signal bandwidths improving the achievable resolution when depicting structures containing materials with ultra-

short T_2^* with stronger read gradients. This enhancement scales with the reduction of the total dead time and the reported speed-up results in an up to 2.75-fold increase in available bandwidth (assuming a 2 μ s excitation pulse, no filter lead-in time, and a 4.5 μ s switching transient [10] compared to 1 μ s using the novel switch). With the realized dead-time, the achievable DC gradient strength becomes the limiting factor for imaging with even higher bandwidth. This in particular if the required field of view is reduced by employing coils with tolerable spurious signal content. This enables acquiring images from compounds with even shorter life-times with higher effective resolution.

Generally, the design of fast T/R switches at lower frequencies of operation is expected to be substantially more difficult since PIN diodes with longer life-times are required for keeping the linearity requirements which increases the accumulated charge during transmission. Furthermore, the spectrum of the switching transients resides closer to the RF carrier or even overlaps which hampers the typically applied frequency-selective filtering of the video leakage signal. Here, the proposed design offers the unique feature that the video leakage is isolated by the different symmetries of the signals and not only by their frequencies. Furthermore, the increase in inductance on the biasing chokes required by a lower frequency of operation would not directly lead to a reduction of the bandwidth and the peak current of the bias in the novel design.

Typical detuning and switched trapping circuits used mutually to decouple receive-only coils and arrays thereof from the volume coil during transmission employ PIN diode switched, resonant tank circuits [40, 41] of basically the same topology as the isolation stages in the presented T/R switch. Therefore the considerations on switching speed, spurious transient signals and provided linearity apply equally in this case. Consequently, the presented PIN diode topology and their biasing can be equally applied for acceleration of switching and dead-time reduction between transmit and receive operation of receive only coils and arrays. For systems employing differential RF signal transmission, biasing of PIN diodes in parallel should offer an equally advantageous isolation of the video leakage by analogously differentiating the signals by their modal structure.

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Tables

Table 1: List of employed testing devices

Device	Type
MR scanner	7T Achieva, Philips Healthcare, Best, Netherlands
Quadrature volume coil	Nova Medical, Wilmington, USA
Acquisition system	MR Acquisition System, Skope Magnetic Technologies Inc, Zurich, Switzerland
Signal generator	SMB100A, Rohde & Schwarz, Munich, Germany
Spectrum analyzer	R&S FSL-3, Rohde & Schwarz, Munich, Germany
Mixed signal oscilloscope	MSO7104B, Agilent, Santa Clara, USA
Vector network analyzer	ENA E5061B, Agilent, Santa Clara, USA
Power amplifier 1 kW	BLA-1000, Bruker BioSpin, Wissembourg, France
Power amplifier 4 kW (2 kW available at coil port)	CPC 9T4000M, Hauppauge, USA
Step attenuator	AS113-50-1, ATM, Patchogue, USA
Directional coupler, -20 dB	Model 3020, Narda, Hauppauge, USA

Table 2: Definitions of measured bias voltage transients

Name	Definition	Description
U_1^{Bias}	$= U_N^{RF} - U_P^{RF\ GND}$	Bias diodes positive side
U_2^{Bias}	$= U_N^{RF\ GND} - U_P^{RF}$	Bias diodes negative side

$U_{RF\ signal}^{CM}$	$= U_N^{RF} + U_P^{RF}$	Common mode voltage on RF signal lines
$U_{RF\ GND}^{CM}$	$= U_N^{RF\ GND} + U_P^{RF\ GND}$	Common mode voltage on RF ground lines
U_{RF}^{SE}	$= (U_P^{RF} + U_N^{RF}) - (U_N^{RF\ GND} + U_P^{RF\ GND})$	Bias voltage in RF signal mode
U_i^{MLF} $i = 1 \dots 4$	Voltage drop over bias resistor	Bias current monitor for monitoring bias currents
U_i^{MLF} $i = 5,6$	Differential output voltage of PIN diode driver	Monitoring of bias voltages

Table 3: Measured S-parameters at 298 MHz

TX state				RX state			
$S_{TX,TR}$ [dB]	$S_{TR,RX}$ [dB]	Z_{TR} [Ω]	Z_{TX} [Ω]	$S_{TX,TR}$ [dB]	$S_{TR,RX}$ [dB]	Z_{TR} [Ω]	Z_{RX} [Ω]
-0.42	-75.6	50.7-0.8i	48.8-2.5i	-41.7	-0.5	49.2-0.3i	48.1-0.8i

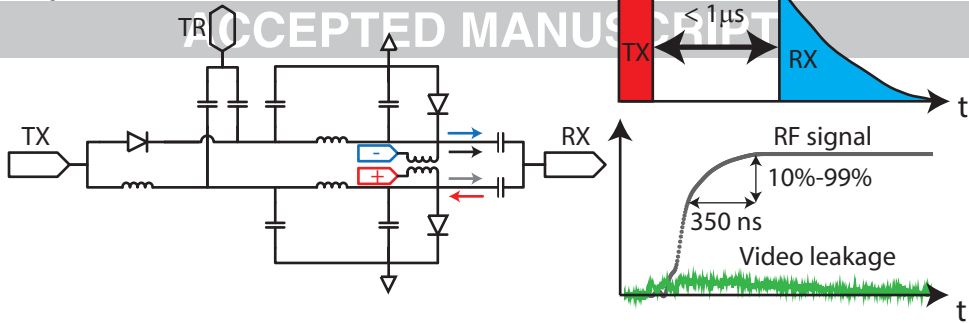
Figure legends

- Figure 1: a) Basic schematic of a PIN diode T/R-switch with low bias voltage topology connecting the coil port (TR) either to the transmitter (TX) or to the receiver (RX). The bias voltage transient is however added to the RF signals creating a large video leakage signal when switching fast. Furthermore, the slewing of the bias voltage is hampered by the bias choke required to block the RF signals from the diode driver. b) Novel approach using differential diode biasing and single ended RF signals. This inherently cancels the video leakage on the RF ports and allows using common-mode chokes for choking the RF signal flowing in common-mode on the two centre lines without introducing a large inductance on the biasing signal pathways running in the differential mode to the PIN diodes through the transformers. 6
- Figure 2: Schematic of the RF signal paths of the proposed T/R-switch. The differential ports UP, NRF, UP, NRF GND, are each connected to a common mode choke for isolating the RF signals from the bias lines. 8
- Figure 3: a) Simplified schematic of the feedback-regulated, high peak-current, differential PIN diode driver employed. b) Block diagram of the malfunction detection circuit implemented to monitor the biasing currents and voltages of the diodes to determine the integrity of the diodes in real time during operation. 9
- Figure 4: a) Block diagram of the integration of 8 T/R-switch modules for driving parallel transmit arrays, including the supply and scanner interface module located outside the Faraday cage. A motherboard stabilizes and distributes the supply voltages and trigger signals for synchronously throwing the switches triggered via a fiber optic interface. The malfunction signals of occupied slots are logically combined to a single optical output (MLF out) indicating that all switches have safely latched the requested state. b) Photograph of the switching unit hosting 2 T/R switch boards. 11
- Figure 5: Bench results. a) Transient voltages from switching of the input trigger (Trig in), the differential bias voltages over the PIN diodes ($U1Bias$, $U2Bias$) and measured at the RX_{Test} port without preamplifier (U_{RX}). b) Voltage transients measured with preamplifier (RX_{LNA}). Furthermore, the in-band (298 MHz) signal rise time and RF leakage are plotted. 15
- Figure 6: Amplitude compression measurements of the power amplifier alone (PA) and of the T/R-switch correcting for the compression of the amplifier. IL = insertion loss. 16
- Figure 7: Test measurements of the switch in the 7T MR scanner. a) FIDs acquired from a rubber mat ($T2 \approx 220 \mu s$) using the proton-free loop coil (blue), as well as mineral oil (red) and a human head (green) using the volume coil. The time $t = 0$ corresponds to the RF pulse center. b)-d) False color images acquired using ZTE with the volume coil. The images are scaled in order to assess the clarity of the background signal. FOV = 530 mm, matrix = 128. b) and c) were acquired with the proposed switches with 250 kHz and 500 kHz bandwidth, respectively. d) was acquired at 500 kHz with traditionally

designed TR switches offering a signal transient time of about 3.5 μ s. e) False color image of the same oil bottle at 500 kHz using the low proton content surface coil. FOV = 340 mm, matrix = 128. f) Image of a rubber mat using the surface-coil setup. FOV = 240 mm, matrix = 128. g) ZTE image of a human head in the volume resonator using 500 kHz bandwidth. FOV = 530 mm, matrix = 320. All images were acquired with a repetition time of 0.5 ms and full angular Nyquist encoding, resulting in scan times of 51 s and 2 m 31 s for the 128 and 320 matrices, respectively. 17

Graphical Abstract

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Highlights

- A novel PIN diode based T/R switch topology is introduced.
- Switching below 1 μs is demonstrated by bench and MRI measurements.
- Very low spurious transient signals are achieved.
- Enhanced switching performance is achieved with state-of-the art signal efficiencies and linearity.